

GOMAL UNIVERSITY DERA ISMAIL KHAN (MAIN CAMPUS)

INSTITUTE OF COMPUTATIONAL INTELLIGENCE BACHELOR OF SCIENCE IN ARTIFICIAL INTELLIGENCE

→ PROJECT. GU_Mini_Robot

→ PROJECT BY. MUHAMMAD DANIYAL

→ SEMESTER. 3rd SEMESTER

→ Roll No. 62413

→ PROGRAM YEAR. 2024 – 2028

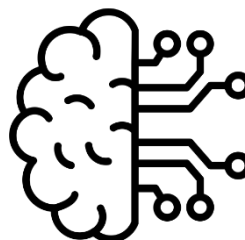
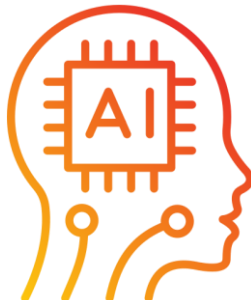
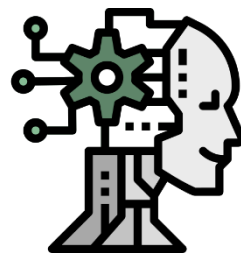
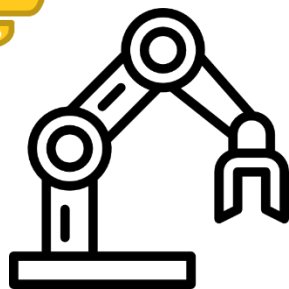
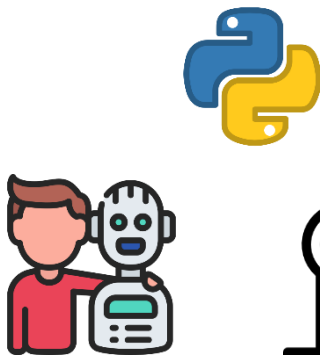


UNDER THE INSTRUCTIONS OF

→ NAME. MS. SHEHR BANO

→ DEPARTMENT. INSTITUTE OF COMPUTATIONAL INTELLIGENCE

→ EXPERT IN. PROGRAMMING FOR ARTIFICIAL INTELLIGENCE
GOMAL UNIVERSITY D.I.KHAN



ARTIFICIAL INTELLIGENCE

Artificial intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.

COMPUTER VISION

Artificial intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.

MACHINE LEARNING

Machine learning (ML) is defined as a discipline of artificial intelligence (AI) that provides machines the ability to automatically learn from data and past experiences to identify patterns and make predictions with minimal human intervention.

DEEP LEARNING

Deep learning is a type of machine learning that uses artificial neural networks to learn from data. Artificial neural networks are inspired by the human brain, and they can be used to solve a wide variety of problems, including image recognition, natural language processing, and speech recognition.

WHAT IS ROBOTICS?

Robotics is a branch of engineering and computer science that involves the conception, design, manufacture and operation of robots. The objective of the robotics field is to create intelligent machines that can assist humans in a variety of ways.

WHAT IS ROBOT?

A robot is a machine capable of carrying out a series of actions automatically, often programmable and designed to perform tasks typically done by humans.

TYPES OF ARTIFICIAL INTELLIGENCE

Narrow AI (Weak AI):

This is the only type of AI that currently exists. It is designed and trained for a specific task, like playing chess, recognizing faces, or powering virtual assistants (Siri, Alexa). While it can perform these tasks very well, it lacks general human cognitive abilities.

Artificial General Intelligence (AGI / Strong AI):

This is a theoretical type of AI that would possess human-like cognitive abilities, capable of understanding, learning, and applying intelligence across a wide range of tasks, much like a human. AGI does not currently exist.

Artificial Superintelligence (ASI):

Also theoretical, ASI would surpass human intelligence in all aspects, including creativity, problem-solving, and general knowledge.

PROGRAMMING LANGUAGES FOR AI

The primary programming language for Artificial Intelligence (AI) is Python, largely due to its simplicity, extensive libraries (like TensorFlow and PyTorch), and large community support.

However, other languages are also crucial for specific AI applications:

<-- R -->

Excellent for statistical analysis and data visualization.

<-- Java -->

Used for building robust, scalable enterprise-level AI applications and integrating AI into existing business systems.

<-- C++ -->

Ideal for performance-critical AI, such as robotics, autonomous vehicles, and game AI, where speed and low-level control are paramount.

<-- Julia -->

Gaining popularity for its high performance in numerical and scientific computing, combining C++'s speed with Python's ease of use.



PYTHON LIBRARIES FOR AI

Machine Learning:

1. **scikit-learn:**

Comprehensive library for machine learning, including classification, regression, clustering, and more.

2. **TensorFlow:**

Open-source library for deep learning, developed by Google.

3. **Keras:**

High-level neural networks API, can run on top of TensorFlow, CNTK, or Theano.

Deep Learning:

i. **PyTorch:**

Dynamic computation graph library for rapid prototyping and research.

ii. **MXNet:**

Scalable and efficient deep learning library.

Natural Language Processing (NLP):

a. **NLTK (Natural Language Toolkit):**

Library for NLP tasks, including text processing, tokenization, and corpora management.

b. **spaCy:**

Modern NLP library for text processing, entity recognition, and language modeling.

c. **Gensim:**

Library for topic modeling and document similarity analysis.

Computer Vision:

1. **OpenCV:**

Comprehensive library for computer vision, including image processing, feature detection, and object recognition.

2. **Pillow:**

Library for image processing and manipulation.



THE PROJECT (Mini_Robot)

HARDWARE SECTION

ARDUINO (The Brain)

Arduino is an open-source microcontroller platform that allows users to create interactive electronic projects, including computer vision applications, by connecting sensors, cameras, and other devices to perform tasks such as image processing, object detection, and robotics.

CORE COMPONENTS OF ARDUINO:

Arduino Board (Hardware)

- This is the physical circuit board containing a microcontroller, which is essentially a small computer.
- It has various input/output (I/O) pins that allow you to connect sensors, actuators (like motors), and other electronic components.
- It also includes a USB port for connecting to a computer and powering the board.

Arduino IDE (Software)

- This is the Integrated Development Environment, a software application that runs on your computer.
- It's where you write code (called "sketches") using a simplified version of the C++ programming language.
- A small program pre-installed on the microcontroller that allows you to upload new code without needing a separate programmer.

THE PROCESS

1. Input:

- Arduino boards can receive input from various sources, such as:
- Sensors (e.g., temperature sensors, light sensors, buttons).
- User input (e.g., through buttons or potentiometers).
- Data from other devices or the internet.

2. Processing:

- The microcontroller on the Arduino board runs the code you've uploaded.
- This code processes the input signals and determines the appropriate output.

3. Output:

- Based on the processed input, the Arduino board can control various output devices, such as:
- LEDs (to turn them on or off, or change their brightness).
- Motors (to control their speed and direction).
- Speakers (to produce sounds).
- Other electronic devices.

4. Programming:

- The Arduino IDE is used to write the code that the microcontroller uses. The code is written in a simplified C++ language.
- The code is then uploaded to the Arduino board via a USB cable.

5. Interaction:

- This process of input, processing, and output allows the Arduino to interact with its environment and perform various tasks.

KEY FEATURES OF ARDUINO

→ Open-source:

The Arduino hardware and software are open-source, meaning that the designs and code are freely available.

→ Ease of use:

Arduino is designed to be easy to learn and use, even for beginners with no prior electronics or programming experience.

→ Versatility:

Arduino can be used for a wide range of projects, from simple LED circuits to complex robotics and automation systems.

→ Community Support:

There is a large and active online community of Arduino users, which provides a wealth of resources and support.

How It Works

The L298N motor driver works by using two H-Bridge circuits to control the direction and speed of DC motors. It allows a microcontroller (like Arduino or Raspberry Pi) to control motors with higher voltages and currents than the microcontroller can handle directly.

Basic Working Principle:

An H-Bridge is an electronic circuit that lets you:

- Apply voltage in either direction across a load (like a motor).
- Control whether the motor spins forward, backward, or stops.

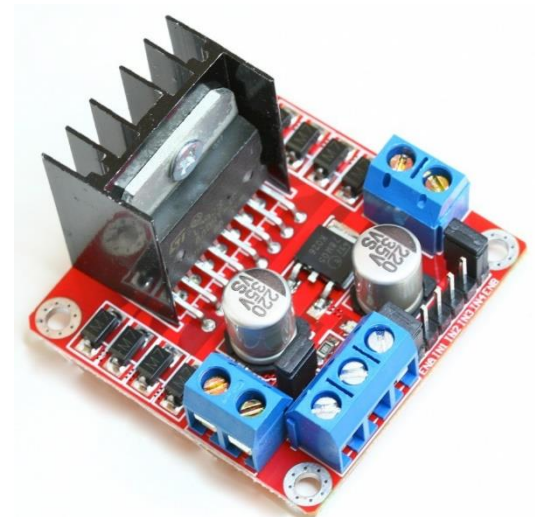
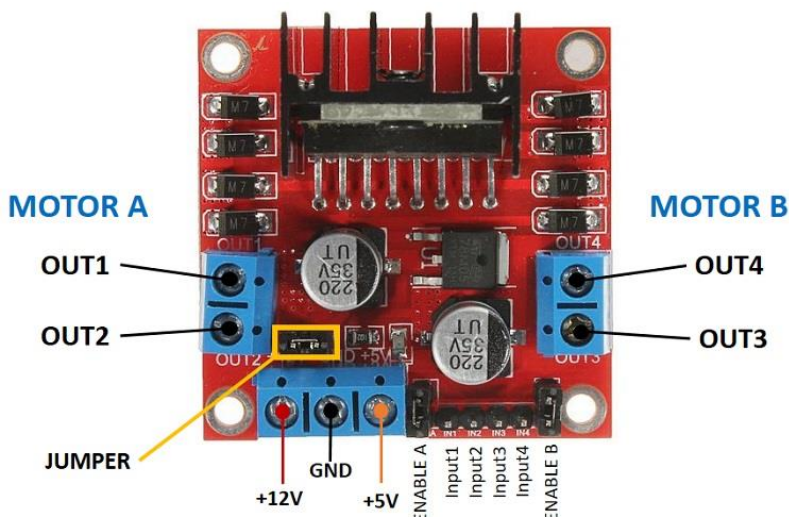
The L298N contains two H-Bridges, so it can control two DC motors independently.

To control direction:

- Set IN1 high, IN2 low → Motor A moves forward
- Set IN1 low, IN2 high → Motor A moves backward

To control speed:

Send PWM signal to ENA (Motor A) or ENB (Motor B) using analogWrite from an Arduino or other microcontroller.



L293D Motor Driver (For DC Motors)

The L293D Motor Driver Module is a compact, easy-to-use device based on the L293D IC, which is a dual H-Bridge motor driver. It's commonly used in DIY electronics projects for controlling two DC motors or one stepper motor with a microcontroller like Arduino, Raspberry Pi, or ESP32.

Key Features of L293D Module

- ➔ Controls 2 DC motors or 1 stepper motor using dual H-Bridge configuration.
- ➔ Supports motor voltages from 4.5V to 36V and operates at 5V logic level.
- ➔ Delivers up to 600 mA per channel with 1.2 A peak current.
- ➔ Allows direction control via input pins and speed control using PWM on ENA/ENB.
- ➔ Includes built-in flyback diodes for protection against motor back EMF.
- ➔ Compact, easy to use, and compatible with Arduino, Raspberry Pi, and other microcontrollers.

Typical L293D Module Pinout

Pin Name	Function
VCC	Logic voltage (typically 5V)
VCC2	Motor power supply (e.g., 6V–12V)
GND	Common ground
IN1/IN2	Control Motor A (direction)
IN3/IN4	Control Motor B (direction)
ENA	Enable Motor A (can use PWM)
ENB	Enable Motor B (can use PWM)
OUT1/OUT2	Motor A outputs
OUT3/OUT4	Motor B outputs

How It Works

The L293D motor driver module works by using dual H-Bridge circuits to control the direction and speed of two DC motors (or one stepper motor), allowing a low-power microcontroller (like Arduino) to drive higher-power motors.

Basic Principle of L293D Module:

- The L293D chip has two H-Bridge circuits, each controlling a motor.
- An H-Bridge allows current to flow in both directions, letting the motor spin forward or backward.

Direction Control:

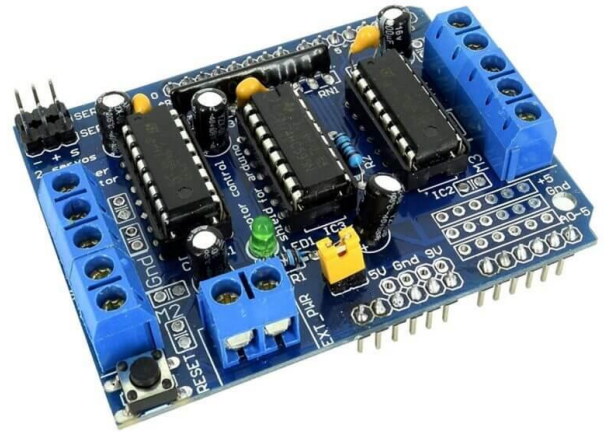
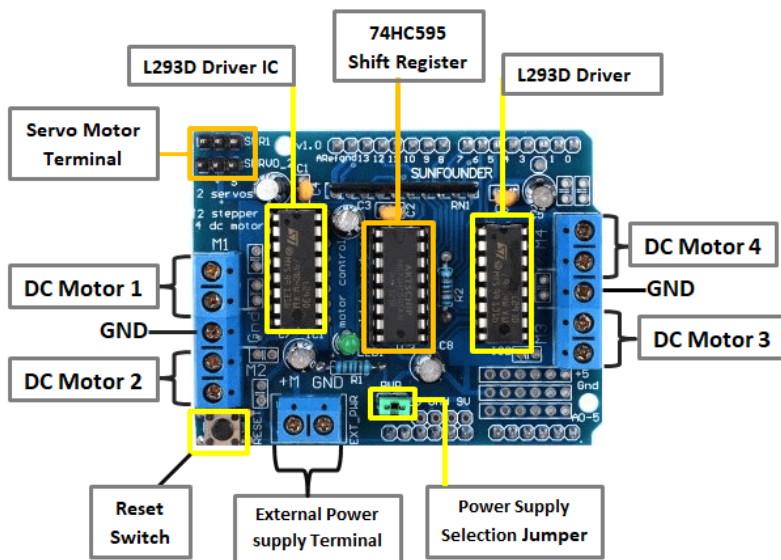
Use the INx pins (e.g., IN1 and IN2) to set the direction of motor rotation.

Example:

- IN1 = HIGH, IN2 = LOW → Motor A rotates forward.
- IN1 = LOW, IN2 = HIGH → Motor A rotates backward.

Speed Control:

- Apply a PWM signal to ENA/ENB to vary speed.
- Use analogWrite() in Arduino to send PWM.



HC-05 Bluetooth Module

The HC-05 is a popular Bluetooth module used for wireless communication in embedded systems and DIY electronics projects. It's widely used for serial communication (UART) between microcontrollers (like Arduino, ESP8266, ESP32) and smartphones/PCs.

Key Features of HC-05

- ➔ Bluetooth 2.0+EDR with UART serial communication at default 9600 baud rate.
- ➔ Operates on 3.3V-5V logic with 10m wireless range in open space.
- ➔ Configurable as Master or Slave device using AT commands.
- ➔ Default pairing PIN "1234" with 2.4GHz frequency operation.
- ➔ Supports baud rate adjustment from 1200 to 1382400 via AT commands.
- ➔ Features status indicator pin (STATE) for connection monitoring.

Pinout of HC-05:

Pin Name	Function
VCC	3.6V - 6V (typically 5V)
GND	Ground
TXD	Transmit (connects to RX of MCU)
RXD	Receive (connects to TX of MCU)
EN/KEY	Used for AT command mode (HIGH = AT mode)
STATE	Indicates connection status (HIGH when paired)

How to Connect HC-05 to Arduino

HC-05	to	Arduino
VCC	→	5V
GND	→	GND
TXD	→	RX (Pin 10)
RXD	→	TX (Pin 11)
EN/KEY	→	(Optional, for AT mode)

How It Works

The HC-05 Bluetooth module works as a wireless serial (UART) bridge, enabling communication between microcontrollers (like Arduino) and Bluetooth devices (smartphones, PCs).

Basic Working Principle:

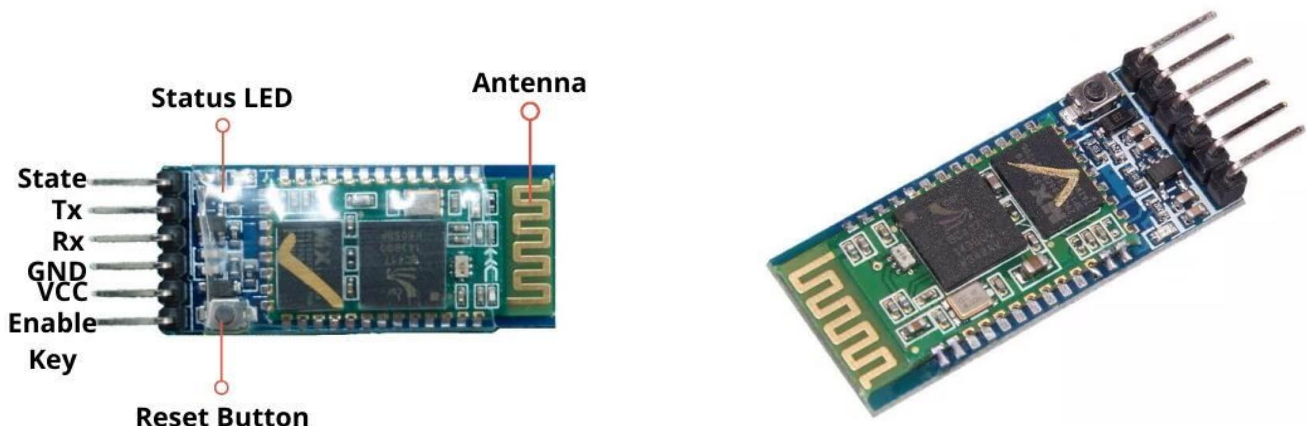
- The HC-05 acts as a wireless replacement for a wired serial connection.
- It transmits data between a microcontroller's UART pins (TX/RX) and a paired Bluetooth device (phone/PC).
- When data is sent from an Arduino, the HC-05 converts it to Bluetooth signals and transmits wirelessly.

Communication Flow:

- Transmit (TX): Data from Arduino → HC-05 → Bluetooth → Phone
- Receive (RX): Data from Phone → Bluetooth → HC-05 → Arduino

Pairing & Connection:

- The HC-05 broadcasts itself (default name: "HC-05").
- A device (phone/laptop) pairs using the default PIN ("1234").
- Once paired, it establishes an RFCOMM serial link (like a virtual COM port).



Jumper Wires

Jumper wires are flexible electrical cables with pre-attached connectors (usually male or female pins) at both ends, used to make quick, temporary connections between electronic components, circuit boards (like Arduino), and breadboards. They serve as the "bridges" in prototyping circuits without soldering.

Key Characteristics:

Connector Types:

- ➔ Male-to-Male (M-M) – Pins on both ends (for breadboard or male-header connections).
- ➔ Male-to-Female (M-F) – Pin on one end, socket on the other (e.g., Arduino to sensor).
- ➔ Female-to-Female (F-F) – Sockets on both ends (e.g., module-to-module linking).

Wire Types:

- Solid-core – Stiff, holds shape in breadboards.
- Stranded – Flexible, better for movable setups.

Common Uses:

- Prototyping circuits on breadboards.
- Connecting microcontrollers (Arduino, ESP32) to sensors, displays, or modules (like HC-05 Bluetooth).
- Testing and debugging electronics.

Standard Specs:

- Spacing: 0.1" (2.54mm) – Matches breadboard and IC spacing.
- Length: 10cm to 50cm (varies by need).
- Color Coding: Red (VCC/+5V), Black (GND), etc., for easy identification.

Why Use Jumper Wires?

- ❖ No soldering needed – Ideal for quick experiments.
- ❖ Reusable – Easily reconfigure circuits.
- ❖ Organized – Color-coding reduces errors.



Robot Chassis

A robot chassis is the physical frame or structure of a robot, serving as the foundational base on which all other components are mounted. It plays a critical role in both the mechanical integrity and functionality of a robot.

Key Components of a Robot Chassis:

- **Base Frame** – The main body that supports all other components.
- **Wheels/Tracks/Legs** – Depending on the locomotion system (e.g., differential drive, omnidirectional wheels, tank treads, or legs for walking robots).
- **Motors & Actuators** – For movement and manipulation (e.g., DC motors, stepper motors, servos).
- **Mounting Points** – For sensors (LiDAR, ultrasonic, IR), controllers (Arduino, Raspberry Pi, STM32), and power systems (batteries).
- **Protective Covering** – Shields delicate parts from dust, water, or impacts (important for outdoor or industrial robots).

Types of Chassis:

Wheeled Chassis: Most common for mobile robots; 2WD or 4WD.

Tracked Chassis: Better for rough terrain; more stable.

Legged Chassis: Mimics animal movement; complex but agile.

Custom Chassis: Tailored for specific applications (e.g., drones, underwater bots).

Materials Used in Robot Chassis:

- + Plastic (ABS, PVC, 3D-printed PLA/PETG) – Lightweight, easy to prototype.
- + Aluminum – Strong, lightweight, corrosion-resistant (common in industrial robots).
- + Steel – Heavy-duty but adds weight (used in large robots).
- + Carbon Fiber – High strength-to-weight ratio (used in advanced robotics).
- + Wood/Acrylic – Common in hobbyist and educational robots.



Components With Robot Chassis

Yellow DC Gear Motors

Yellow DC Gear Motors are a very common type of motor used in beginner and intermediate robotics projects. They're inexpensive, easy to use, and widely supported in DIY kits.

Key Features of Yellow DC Gear Motors:

- ➔ Voltage Range: Usually 3V to 12V (common: 6V or 12V).
- ➔ Speed: ~100 to 300 RPM (varies based on gear ratio).
- ➔ Torque: ~1 to 10 kg·cm (enough for small to medium robots).
- ➔ Shaft Type: D-shaped or round shaft (for easy wheel/mounting).
- ➔ Built-in Gearbox: Often plastic or metal gears (metal lasts longer).
- ➔ Encoder Option: Some models include encoders for speed/position feedback (useful for PID control in robotics).

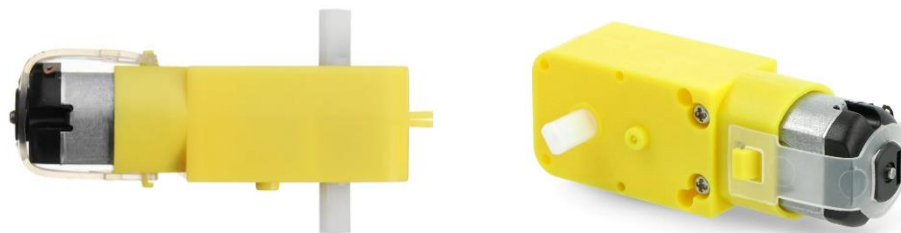
Common Uses in Robotics:

- ❖ Wheeled Robot Chassis (Differential drive, 4WD robots).
- ❖ Robotic Arms & Grippers (for joint movements).
- ❖ Conveyor Belts & Small Automation Systems.
- ❖ Tracked Vehicles (tank-like robots).

Wiring & Control:

These motors are brushed DC, so they can be controlled via:

- ❖ L298N / L293D Motor Driver (for Arduino/RPi).
- ❖ TB6612FNG (more efficient, less heat).
- ❖ Relay or MOSFET (for simple on/off control).



AAA Battery Slot

A AAA battery slot (also known as a battery holder or battery case) is a plastic or metal housing designed to hold one or more AAA batteries securely and connect them to a circuit.

Key Features of a AAA Battery Slot:

Feature	Description
Battery Size	AAA (1.5V each)
Materials	Usually plastic (ABS) with metal (spring/contact terminals)
Number of Cells	1 to 4 AAA cells typically (1.5V to 6V total)
Terminal Types	Bare wire leads, solder tabs, or connectors (e.g., JST)
Mounting Options	Screw holes, adhesive backing, or PCB mounting pins
Polarity Markings	"+" and "-" clearly labeled

Common Uses in Robotics & Electronics:

- ➔ Powering microcontrollers (like Arduino)
- ➔ Driving small DC motors (e.g., yellow gear motors)
- ➔ Portable electronics
- ➔ Educational kits



Connections

HC-05 Bluetooth Module to Arduino

HC-05	to	Arduino
VCC	➔	5V
GND	➔	GND
TXD	➔	RX (Pin 0)
RXD	➔	TX (Pin 1)
EN/KEY	➔	(Optional, for AT mode)

L298N Motor Driver to Arduino

L298N	to	Arduino
GND	→	GND
IN1	→	Pin 2
IN2	→	Pin 3
ENA	→	Pin 9 (PWM)
IN3	→	Pin 4
IN4	→	Pin5
ENB	→	Pin 10 (PWM)

Yellow Gear DC Motors to L298N Motor Driver

DC Motors	to	L298N
Motor A	→	OUT1
Motor A	→	OUT2
Motor B	→	OUT3
Motor B	→	OUT4

AAA Battery Slot to L298N Motor Driver

AAA Battery Slot	to	L298N
Positive (+) Wire	→	12V
Negative (-) Wire	→	GND

Doubling Combination

